🚀 React Training — Day 1

Introduction & First React App

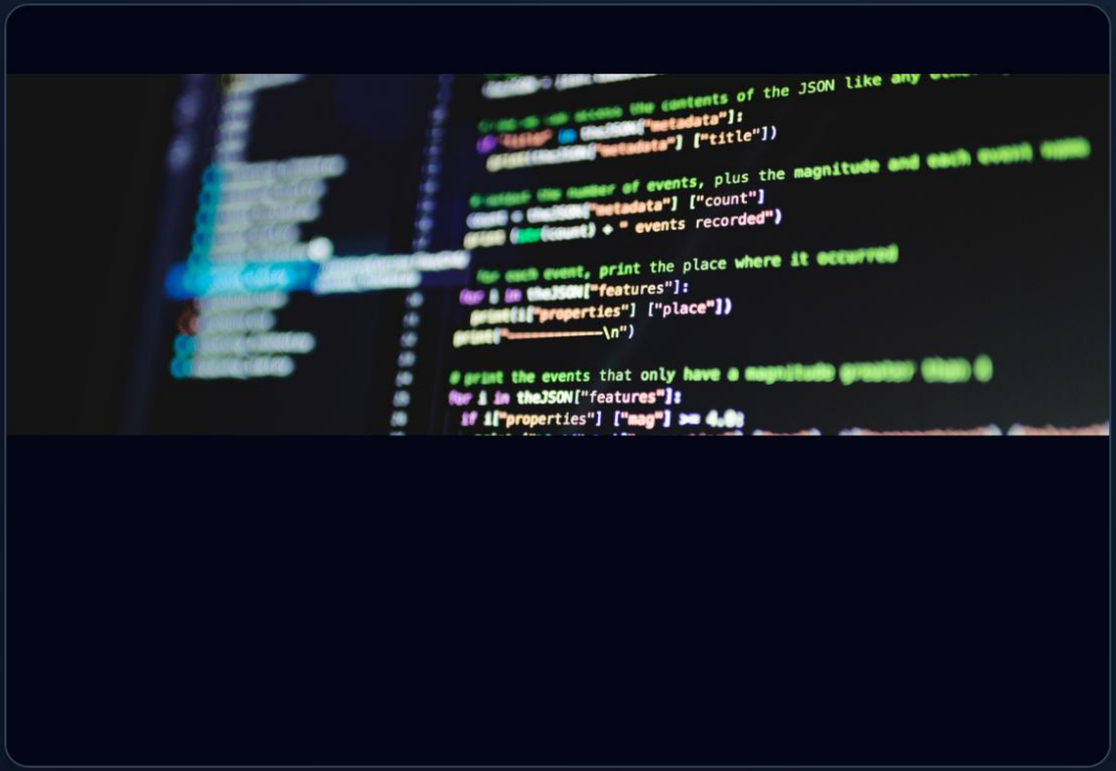
🎯 Goal of Today

Understand React basics and run your first React application.

- **What React is:** Understanding the core concepts and purpose.
- **Why React is popular:** Learn about components and rapid rendering.
- **Install tools:** Setting up your development environment.
- **Create project:** Scaffolding your very first React setup.
- **Write code:** Building and editing your first React UI component.

What is React?

- React is a **JavaScript library** used heavily to build user interfaces (UI).
- Created and maintained actively by **Meta (Facebook)**.
- React is the industry standard, used in major platforms like **Netflix, Instagram, and Airbnb**.
- It provides an architecture that helps developers build modern, extremely **fast web applications**.



Real-Life Example

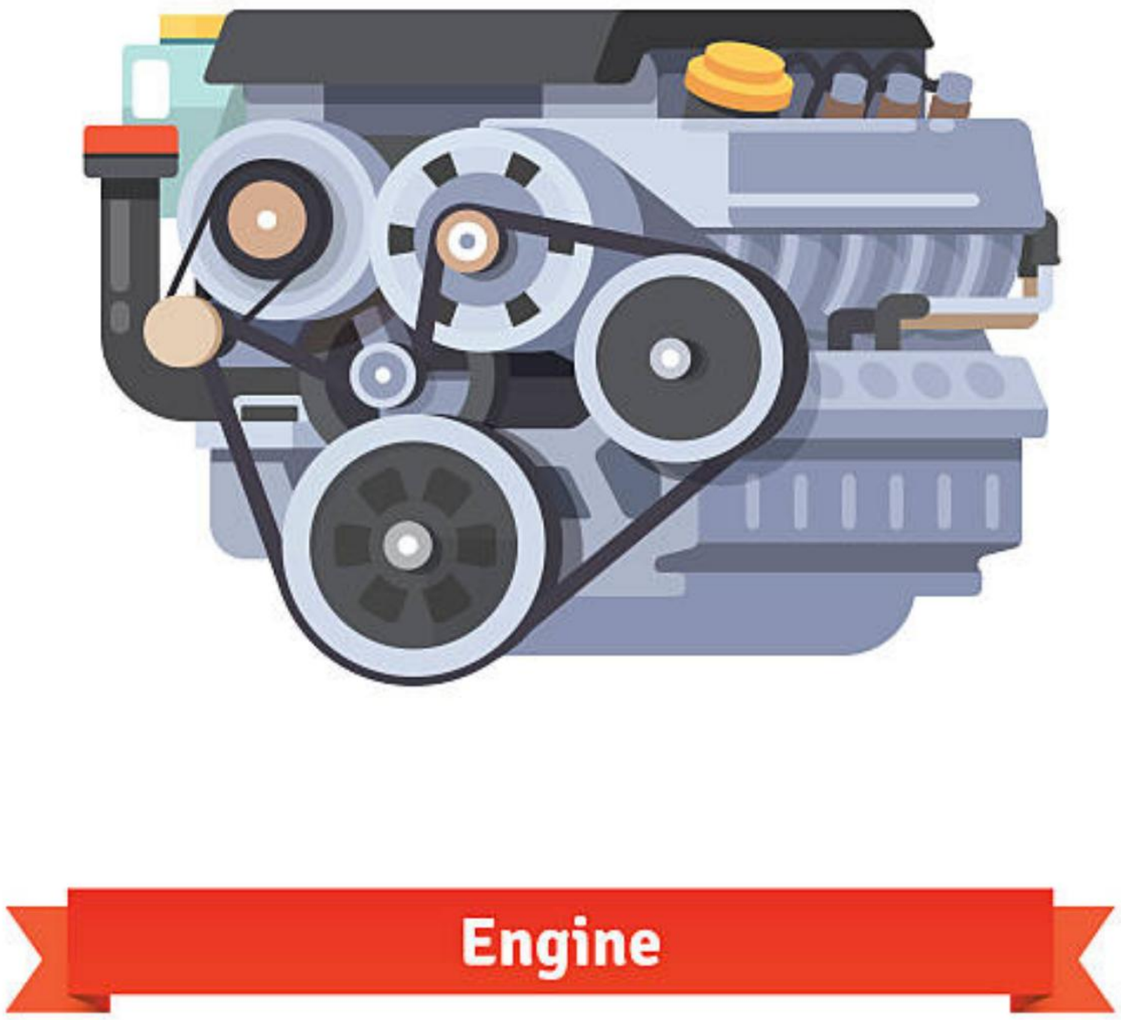
Think of a car 🚗

A car has many distinct, individual parts: **Engine, Wheels, Seats, Doors**.

Each part is manufactured to work independently, but together they form the complete functional vehicle.

React works the exact same way.

Instead of building one massive, tangled page, a website is divided into small, independent, and reusable parts called **Components**.



Example Website Structure

A website built with React is composed entirely of distinct component pieces:

Navbar

The top navigation component, handling app menus and routing links.

Sidebar & Content

The main body components, rendering the heavy content like articles and product cards.

Footer

The bottom layout component containing standard copyrights and secondary links.

Why React is Popular

Component-Based

The UI is architected exactly like **LEGO blocks**.

You create small logic pieces like `Navbar`, `ProductCard`, and `Footer`.

These isolated, highly reusable components save immense development time and minimize bugs.

Fast Rendering

React meticulously updates **only the exact part that changes**.

Example: On YouTube, when you click Like 👍, only the like count text updates.

The whole page does not reload, making the user experience lightning fast.

Reusable Code

Write Once, Use Everywhere

React components are designed to be heavily reused across your application.

Instead of duplicating HTML structure for every single product, you just invoke the same component multiple times.

This drastically reduces duplicate code files and makes maintenance incredibly effortless.

React vs Normal JavaScript

Feature	Normal Website	React Website
Page Action	Whole page continuously reloads	Only the necessary part updates
Code Maintenance	Hard to maintain large spaghetti code	Clean, modular, Component-based
File Structure	Large, monolithic document files	Small, isolated, reusable files

React Prerequisites

Before diving into React, you should be comfortable with:

- **HTML:** Semantic structure of the web page.
- **CSS:** Styling, layouts, and responsiveness.
- **JavaScript (ES6):** Variables, Arrow Functions, and Array maps.

> **Since you already started HTML and JavaScript earlier, you are completely ready!**

Install Required Tools

We need three essential tools for local development:

Node.js

Download from **nodejs.org**. Node allows us to run JS server tools. Verify installation with `node -v`.

VS Code

Download from **code.visualstudio.com**. The industry standard, highly customizable code editor.

Web Browser

Google Chrome, Mozilla Firefox, or Edge to test and view your React applications live.

VS Code Recommendations

Recommended Extensions

- ES7+ React/Redux Snippets
- Prettier - Code formatter
- Live Server

Why use them?

These extensions are critical for modern web development.

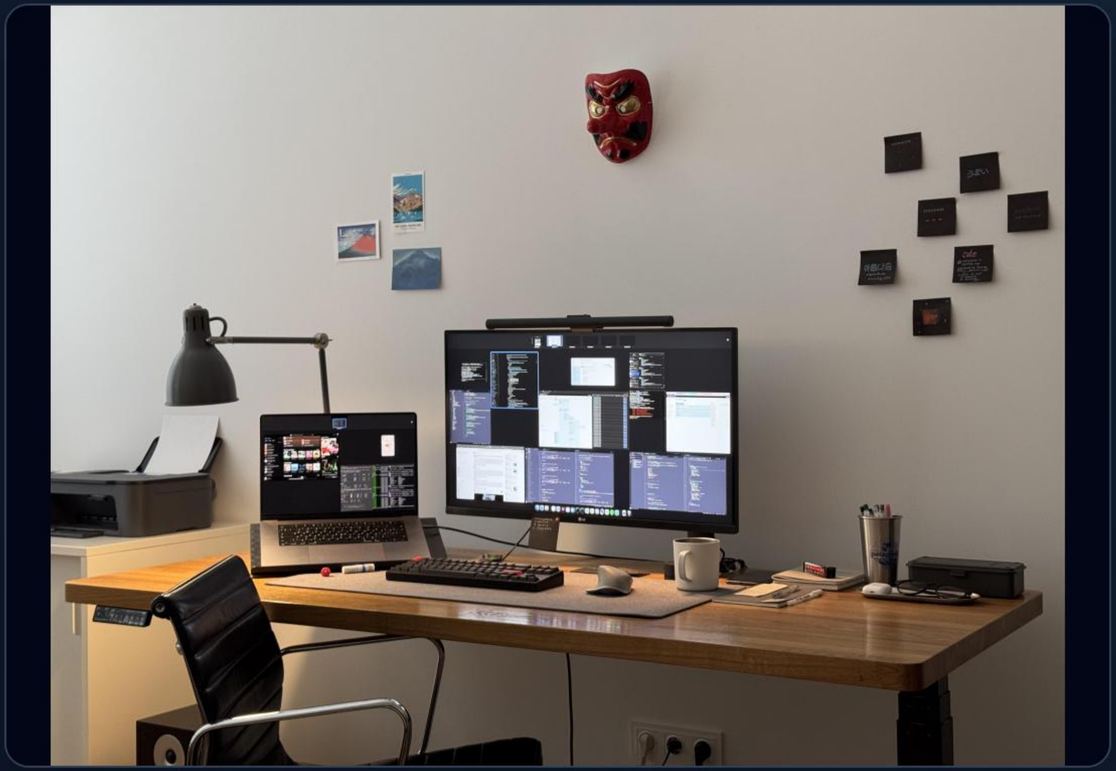
They strictly enforce cleaner code by automating repetitive formatting tasks and providing instant boilerplate code generation shortcuts.

Create First React App

Modern React projects consistently use **Vite** for a lightning-fast build setup.

```
$ npm create vite@latest
```

- **Project name:** `react-learning`
- **Select framework:** `React`
- **Select variant:** `JavaScript`



Run the Project

Navigate to your folder and start the development server:

```
$ cd react-learning
$ npm install
$ npm run dev
```

- You will see a local URL in the terminal like: `http://localhost:5173`
- Open this exact URL in your browser application.

🎉 **Your React app is running!**

React Project Structure

Core Folders

- **node_modules:** Contains all downloaded NPM packages.
- **public:** Static files like favicons that bypass compilation.
- **src:** The source folder where you write 99% of your logic.

Core Files

- **index.html:** The foundational HTML file wrapping the app.
- **main.jsx:** The core entry point of the React application.
- **App.jsx:** The primary root component of your application UI.

Your First React Code

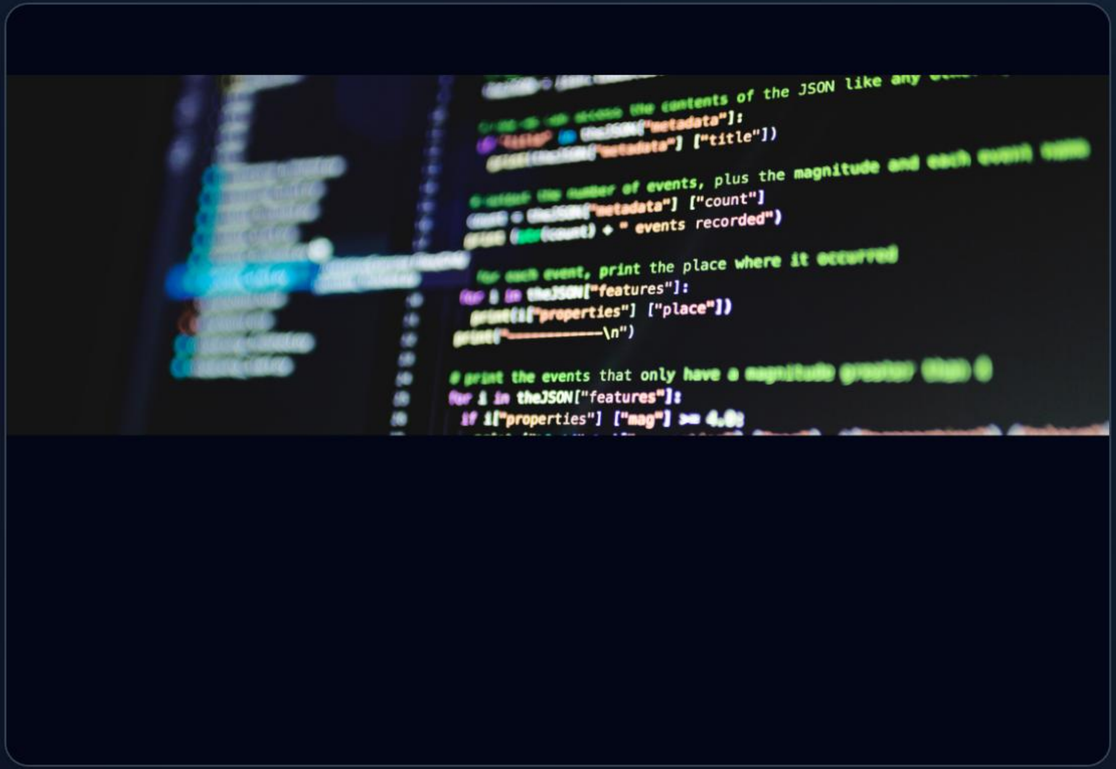
Open `src/App.jsx` and replace everything with this code block:

```
function App() {
  return (

      Hello Saurav 🚀
      My first React app

  )
}

export default App
```



Save the file, and watch the browser!

Hot Reloading

Hot Reloading

Instant Feedback

React, heavily powered by Vite, automatically refreshes the UI the exact moment your code changes and is saved.

This revolutionary productivity feature is called **Hot Reloading**.

You never need to reload the page manually again!

Practice Task

Build Your Profile

Using the `App.jsx` file you just edited, create this UI inside the return block:

- **Title:** My Portfolio
- **Name:** Saurav
- **Role:** Future Full Stack Developer
- **Button:** Hire Me

What You Learned Today

- **Core Concepts:** What React is & Why it is immensely popular.
- **Architecture:** The Component design concept and Code Reusability.
- **Environment Setup:** Installed Node.js & Visual Studio Code.
- **Project Initialization:** Created a React project rapidly with Vite.
- **Execution:** Ran a React app, edited code, and tested Hot Reloading.

Tomorrow (Day 2)

Next Key Topics:

React Folder Structure Deep Dive

JSX Comprehensive Understanding

Building Your First Custom Components

Essential React Rules

You will build: 🧩 React Profile Card

Image Sources

https://welovedevs.com/wp-content/uploads/2024/09/banniere-article-WP-dhiWise.png

Source: welovedevs.com

---

https://media.istockphoto.com/id/492676668/vector/modern-car-internal-combustion-engine.jpg?s=612×612&w=0&k=20&c=W9jM0nNV3OcxnMnqysvunvc55d7252aNn5jlyHPdK8A=

Source: www.istockphoto.com

---

https://i.redd.it/m0o40m8k8jtd1.jpeg

Source: www.reddit.com

---

https://imgv3.fotor.com/images/homepage-feature-card/fotor-3d-avatar.jpg

Source: www.fotor.com

---

https://static.vecteezy.com/system/resources/previews/014/409/275/non_2x/abstract-technology-digital-futuristic-concept-glow-neon-colors-lines-with-lighting-effect-perspective-on-dark-blue-background-vector.jpg

Source: www.vecteezy.com